



HK IMO Training 2021-2022
Problem Set 9
For 27 November 2021



Problem 33. There are 5 sets such that the intersection of any two sets has exactly 1 element, and the intersection of any four sets is empty. Find the smallest possible number of elements in the union of these 5 sets.

Problem 34. Let D be a point on the side BC of $\triangle ABC$. Let E be a point on the internal angle bisector of $\angle BAC$ which lies inside $\triangle ABC$. Suppose AE and BE meet the circumcircle of $\triangle ABD$ again at P and Q respectively (with $P \neq Q$). Suppose AE and CE meet the circumcircle of $\triangle ACD$ again at X and Y respectively (with $X \neq Y$). Prove that BC , PQ and XY are concurrent or parallel.

Problem 35. We choose 42 integers from 1 to 1000000 randomly. Prove that there exist 4 chosen numbers such that for any permutation (a, b, c, d) of them, we have $25(ab + cd)(ac + bd) \geq 16(ad + bc)^2$.

Problem 36. Prove that for any positive integer N , there exists a prime p and a multiplicative subgroup G of $\mathbb{F}_p^\times = \{1, 2, \dots, p-1\}$ such that G has at least N elements, and there are no $x, y, z \in G$ (which can be equal) satisfying $x + y \equiv z \pmod{p}$. (A multiplicative subgroup of $\mathbb{F}_p^\times$ is of the form $\{a^k \pmod{p} : k \geq 0\}$ for some $a \in \mathbb{F}_p^\times$.)